NTIRE2026 – Robust AI-Generated Image Detection in the Wild

Three Guys
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1 Team details

a) Team name - Three Guys

b) Team leader name - Sharan K

c) Team leader institution and email - Shiv Nadar University, Chennai and sharank3026@gmail.com

d) Rest of the team members - Adhithya Srivatsan, Ankith V

e) Team website URL (if any) - N/A

f) Affiliations - Shiv Nadar University Chennai

g) Usernames on the *NTIRE2026 – Robust AI-Generated Image Detection in the Wild* CodaBench leaderboard (development/validation and testing phases) - sharank

h) Link to the codes/executables of the solution(s) - https://github.com/addytrunks/IVP_NTIRE_2026

Three Guys

Title: Robust AI-Generated Image Detection in the Wild using EfficientNet-B5 and FFT
Members:
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2 Contribution details

a) Title of the contribution
Robust AI-Generated Image Detection using a 4-Channel EfficientNet-B5 with GPU-Accelerated Frequency-Domain Fusion and Stochastic Augmentation

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b) Overall approach
The proposed method addresses robust AI-generated image detection under real-world "in-the-wild" degradations such as JPEG compression, blur, noise, and resolution reduction. Beyond standard spatial analysis, the model fuses frequency-domain information to capture periodic synthetic artifacts that persist even after heavy post-processing.

The architecture uses a modified EfficientNet-B5 backbone accepting a 4-channel input tensor (3 RGB + 1 FFT). The frequency branch computes a log-magnitude 2D Fast Fourier Transform (FFT) spectrum on-the-fly on the GPU using the luminosity grayscale map, which is then min-max normalized to produce a single additional channel. The first three input channels are initialized from pretrained ImageNet weights; the fourth (FFT) channel is initialized as the mean of the RGB weights, preserving the Squeeze-and-Excitation blocks' channel-wise attention capability.

To maximize the primary Robust ROC AUC metric, a stochastic distortion pipeline applies 1–3 random transformations per image with 60% probability, sampling severity from a Gaussian distribution centered at mild levels. The seven distortion groups include blur, noise, JPEG compression, color distortions, brightness changes, spatial jitter, and tonal quantization.

Training uses BCEWithLogitsLoss and AdamW with a OneCycleLR scheduler. To fit within the 16 GB VRAM of the Tesla P100, Automatic Mixed Precision (AMP) and gradient accumulation over 8 steps are employed, achieving an effective batch size of 32.

c) References

- M. Tan and Q. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," *ICML*, 2019.
- Ross Wightman, PyTorch Image Models (timm), `https://github.com/huggingface/pytorch-image-models`
- A. Buslaev et al., "Albumentations: Fast and Flexible Image Augmentations," *Information*, 2020.
- NTIRE 2026 Challenge: Robust AI-Generated Image Detection in the Wild.

d) Representative image / diagram of the method
The proposed architecture combines a spatial RGB branch and a GPU-accelerated FFT frequency branch, concatenated into a 4-channel tensor fed into a modified EfficientNet-B5 stem.

e) Have you tested previously published methods (yes/no)?
No.

f) Other methods and baselines tested (even if results were not top competitive).
EfficientNet-B0, EfficientNet-B4, Swin Transformer, and EfficientNet-B5 with PID features and contrastive loss were evaluated as ablation baselines prior to the final submission.

3 Global Method Facts

a) Total method complexity: all stages
The pipeline includes on-the-fly frequency feature computation, a stochastic 12-transformation augmentation stage, feature extraction via a 4-channel EfficientNet-B5 backbone (∼30M parameters), and binary classification. Inference complies with the 1 FPS and 80 GB GPU memory constraints mandated by the challenge.

b) Which pre-trained or external methods/models have been used (for any stage, if any)?
EfficientNet-B5 pretrained on ImageNet, obtained via the timm (PyTorch Image Models) library. The first three input channels of the modified stem are initialized from these pretrained weights.

c) Do you use extra data besides the competition data provided? (yes/no)
No. The model was trained exclusively on the NTIRE 2026 challenge dataset (∼277K images, 20 generative models, 12 training-phase distortions).

d) Training description
Training is performed in PyTorch on a Tesla P100 (16 GB VRAM). Images are resized to 456×456 and normalized using ImageNet statistics. The BCEWithLogitsLoss objective is optimized with AdamW (learning rate 1×10^{-4}, weight decay 1×10^{-2}) under a OneCycleLR schedule. AMP and 8-step gradient accumulation give an effective batch size of 32. The stochastic augmentation pipeline applies distortions at 60% probability per image. Training runs for 4 epochs and the best checkpoint by validation AUC is retained.

e) Testing description
Inference is performed on the best validation checkpoint. The 4-channel input (RGB + FFT) is constructed identically to the training forward pass. No test-time augmentation is applied in the final submission. The model produces a binary probability score for each image indicating whether it is AI-generated.

f) Advantages of the proposed solution

- GPU-accelerated on-the-fly FFT computation adds no preprocessing overhead and stays within latency constraints.
- Frequency features capture periodic upsampling artifacts from generative models that survive spatial degradations.
- Stochastic Gaussian-severity augmentation directly optimizes for the Robust ROC AUC primary metric.
- AMP and gradient accumulation allow training a large backbone on constrained hardware without compromising effective batch size.
- Single-model approach avoids ensemble complexity while remaining competitive.

g) Results of the comparison to other approaches
An ablation study comparing several architectures is summarized in Table 2. The 4-channel EfficientNet-B5 (RGB + FFT) achieved the best trade-off between Clean AUC (0.9768) and Robust AUC (0.9341), outperforming both lighter EfficientNet variants and the Swin Transformer on the robustness metric.

h) Results on other benchmarks
Not evaluated on external benchmarks beyond the NTIRE 2026 leaderboard.
Validation phase (193 participants): Clean ROC AUC 0.9768, Robust ROC AUC 0.9341, Clean Hard ROC AUC 0.9220, Robust Hard ROC AUC 0.8324 – 100th place.
Testing phase (94 active participants): Clean ROC AUC 0.8084, Robust ROC AUC 0.7099 – 75th place.

i) Novelty degree of the solution, and if it has been previously published
The primary novelty lies in the direct concatenation of a GPU-computed log-magnitude FFT channel with the RGB input, enabling the EfficientNet-B5 Squeeze-and-Excitation blocks to perform channel-wise attention across spatial and frequency cues jointly. The stochastic Gaussian-severity augmentation strategy tailored for Robust AUC optimization is an additional practical contribution. This work has not been previously published.

j) **It is OK if the proposed solution is based on other works (papers, reports, Internet sources (links), etc). It is ethically wrong and misconduct if you do not properly give credit and hide this information.**

Please fill in Table 1 specifying the technical information (besides writing it), which should take 1 minute. Ensemble refers to model ensembles and self-ensembles.

Input	Training Time	Epochs	Extra data	Attention	Quantization	Base Model	# Params. (M)	Runtime	GPU
(456, 456, 4)	∼2h	4	No	Yes (SE)	No	EfficientNet-B5 (4-ch)	∼30M	>1 FPS	Tesla P100

Table 1: Overview of method specification.

Model Variant	**Clean AUC**	**Robust AUC**
EfficientNet-B0	0.9290	0.8402
EfficientNet-B4	0.9410	0.9212
Swin Transformer	0.9709	0.8965
EffNet-B5 (RGB + FFT)	**0.9768**	**0.9341**
EffNet-B5 (PID, FFT, Cont. Loss)	0.9506	0.8701

Table 2: Ablation study: performance comparison of model architectures on the validation set.

4 Competition particularities

The NTIRE 2026 Robust AI-Generated Image Detection in the Wild challenge focuses on detecting manipulated images under various image degradations. Our method specifically addresses this requirement by introducing a stochastic degradation-based augmentation pipeline during training with Gaussian-sampled severity levels. By exposing the model to realistic distortions such as blur, compression artifacts, noise, and resolution reduction, the model learns more robust representations and improves its ability to detect AI-generated images in degraded conditions. The dual-domain (RGB + FFT) input further ensures that frequency-level forgery signatures remain detectable even when spatial cues are destroyed by post-processing.

5 Technical details

The solution is implemented in PyTorch using a 4-channel EfficientNet-B5 backbone from the timm library. The initial convolutional stem is replaced with a $4 \to C$ variant; the first three channel weights are copied from the ImageNet-pretrained model and the fourth is set to the mean of the RGB weights.

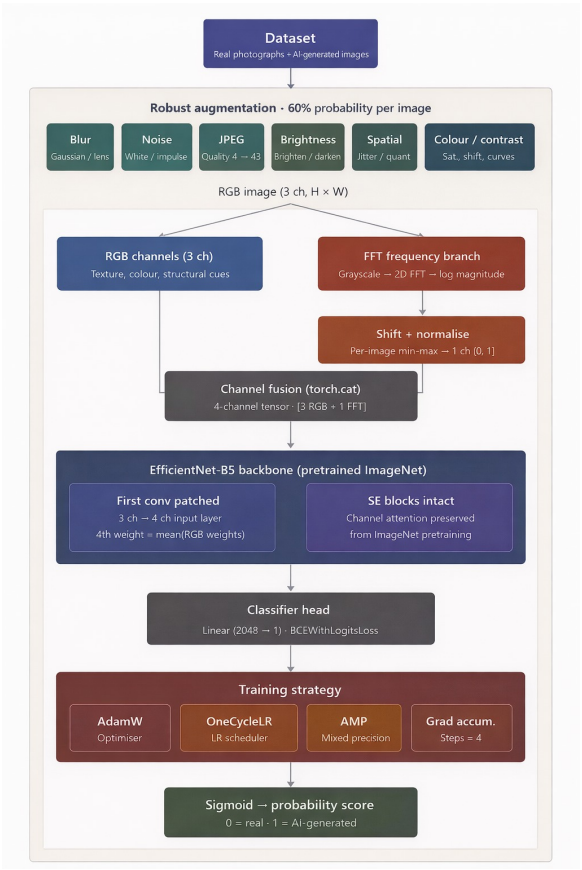


Figure 1: Architecture

The frequency branch computes the grayscale luminosity map of the input, applies a 2D FFT with frequency shift, and extracts the log-magnitude spectrum: $S = \log(|FFTshift(G)| + 10^{-6})$, followed by per-image min-max normalization to $[0, 1]$.

Training uses AdamW (lr $= 1 \times 10^{-4}$, weight decay $= 1 \times 10^{-2}$) with a OneCycleLR scheduler. Automatic Mixed Precision and gradient accumulation (8 steps, effective batch size 32) are employed to fit the Tesla P100 16 GB VRAM budget. The stochastic distortion pipeline selects 1–3 random distortions per image from 7 groups at 60% probability, with severity sampled from a Gaussian centered at mild levels. Training runs for 4 epochs; the checkpoint with the best validation AUC is used for final inference.

6 Other details

a) Planned submission of a solution description paper at NTIRE 2026 workshop
NO

b) General comments and impressions of the *NTIRE2026 – Robust AI-Generated Image Detection in the Wild*

The challenge provides a rigorous benchmark for evaluating AI-generated image detection under highly realistic degradation conditions. The emphasis on Robust ROC AUC as the primary metric is well-motivated and pushes participants toward methods that generalize beyond clean, ideal inputs. The progressive increase in distortion complexity from training (12 transforms) to testing (24 transforms) effectively stress-tests model robustness.

c) What do you expect from a new challenge in AI-generated image detection?
Future iterations could benefit from larger and more diverse training sets spanning a wider range of generative architectures, cross-domain evaluation tracks, and the provision of cloud GPU credits to reduce the hardware barrier for academic teams.